

Volcano社区贡献指南

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01

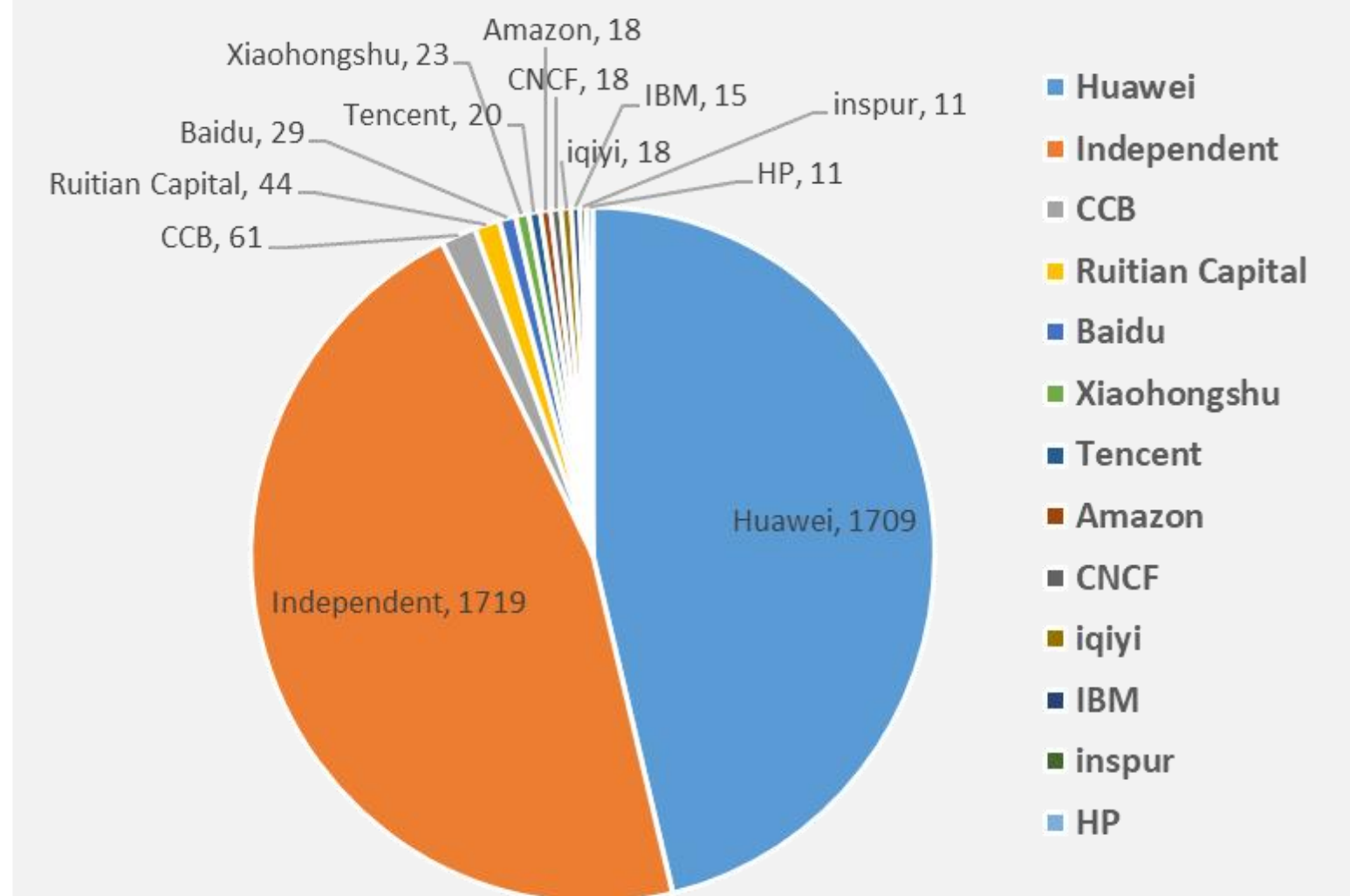
社区生态现状、规划



Volcano 社区开发者生态

- CNCF Incubating project: <https://landscape.cncf.io/selected=volcano>
- 2.6k+ GitHub Stars
- 150+ Contributors from Huawei, AWS, Google, Facebook, RedHat, Oracle, OpenAI, Baidu, Tencent, BoCloud, Bytedance, Ximalaya, BiliBili, Ruitian Capital, 360, etc.
- 22+ formal releases
- 50+ public adopters (private not included)
- CII best practice: passing
- License: Apache 2 License

Volcano 社区多样性



Data from <https://volcano.devstats.cncf.io>



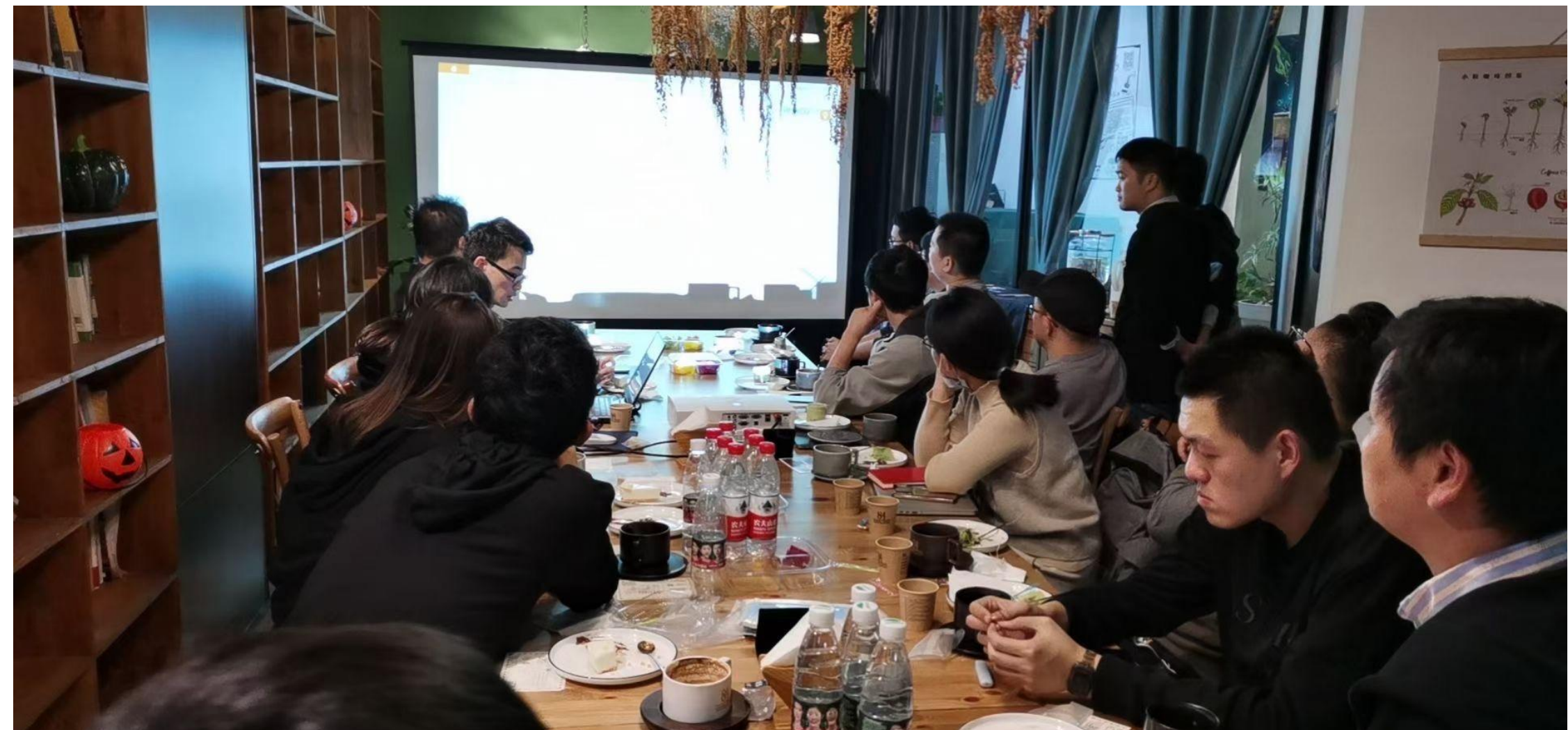
Volcano 社区用户生态



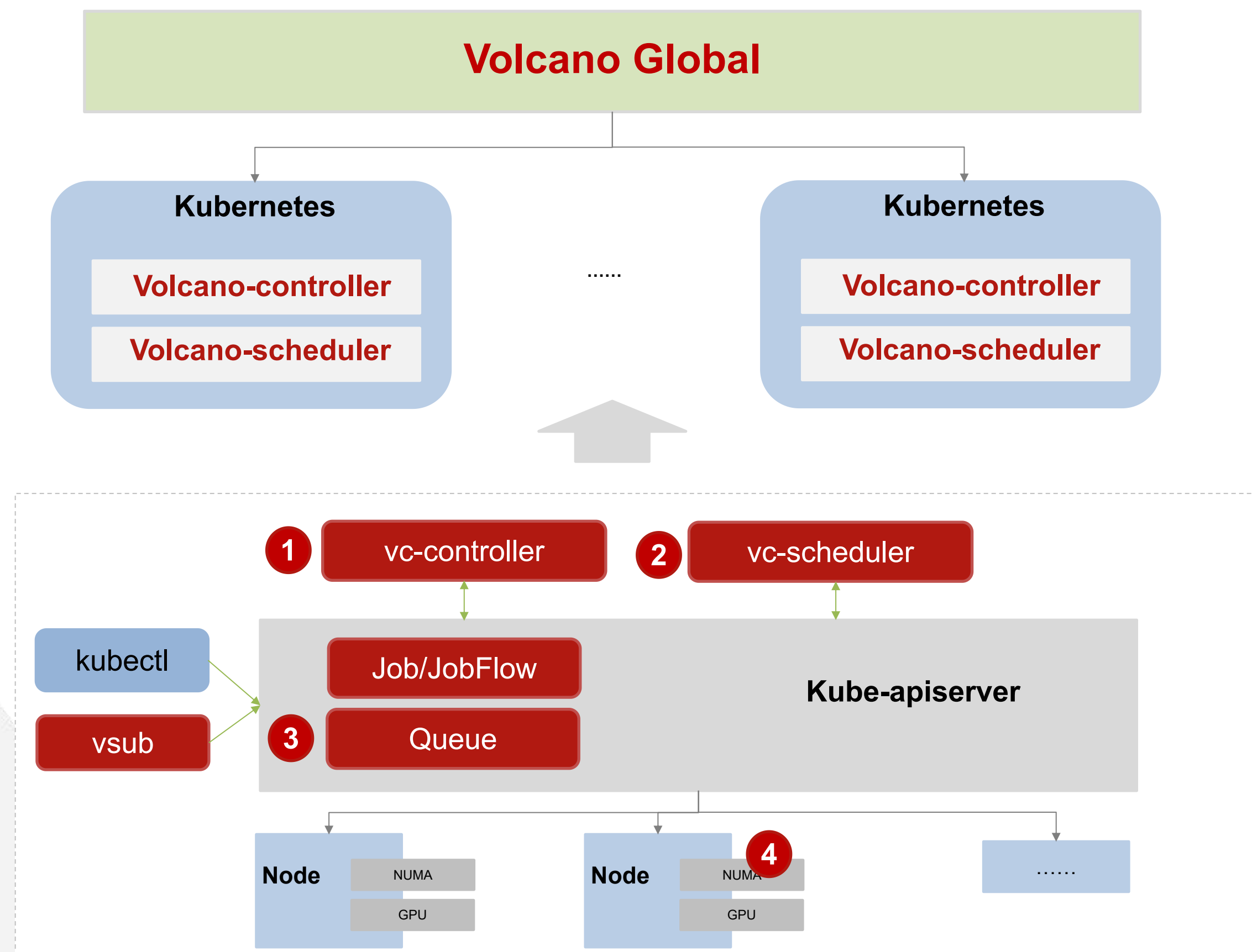
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Volcano 社区活跃度



Volcano 总览



1. 统一的作业管理

提供完善作业生命周期管理，统一支持几乎所有主流的计算框架，如Pytorch, MPI, Horovod, Tensorflow、Spark-operator, Flink-operator等。

2. 丰富的高阶调度策略

公平调度、任务拓扑调度、基于SLA调度、作业抢占、回填、弹性调度、混部等。

3. 细粒度的资源管理

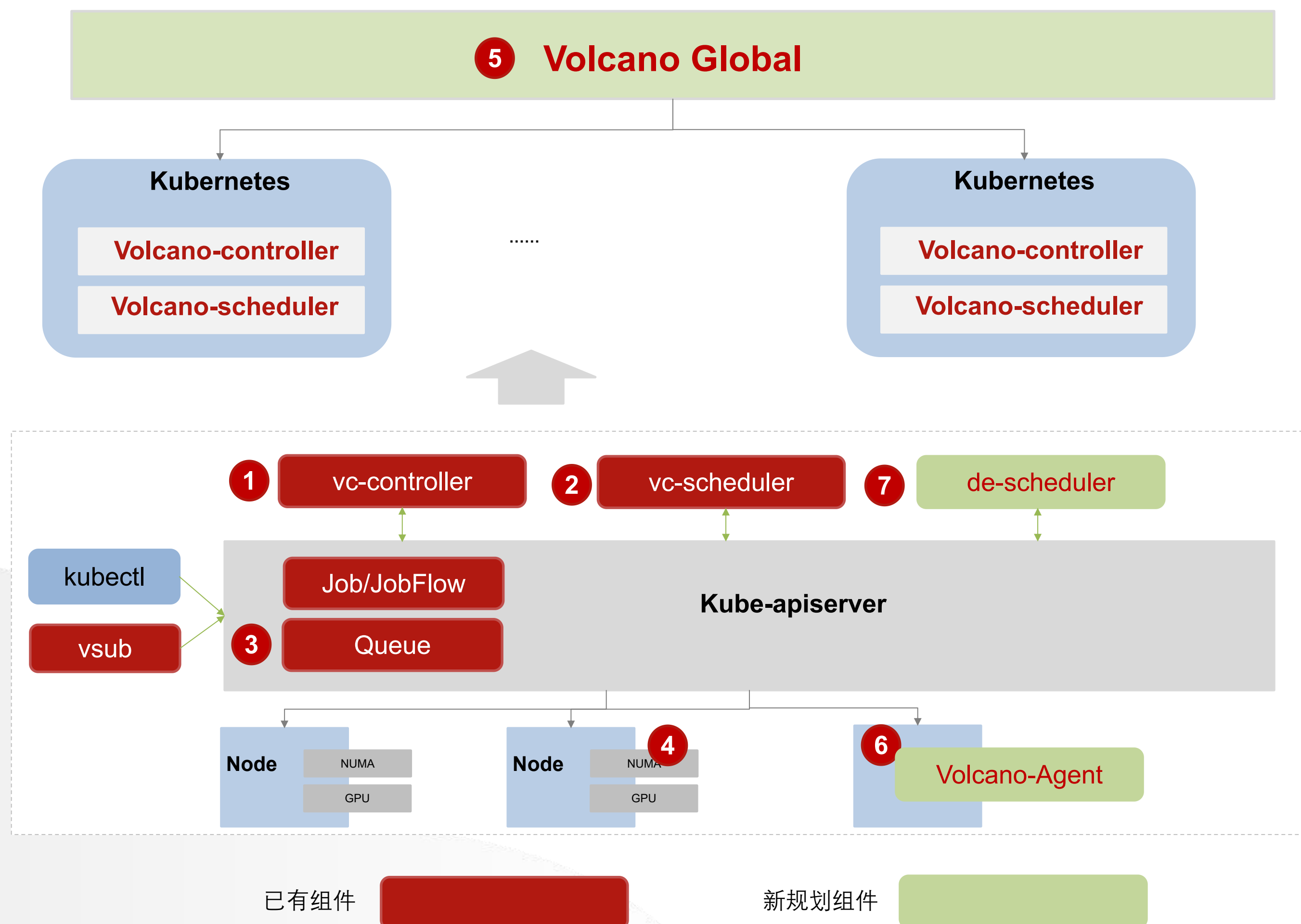
提供作业队列，队列资源预留、队列容量管理、多租户的动态资源共享。

4. 性能优化和异构资源管理

调度性能优化，并结合 Kubernetes 提供扩展性、吞吐、网络、运行时的多项优化，异构硬件支持x86, Arm, GPU, 昇腾, 昆仑等。



Volcano 未来规划



1. 多云全域调度

- 多租公平、全局资源最优

2. FinOps成本洞察优化

- 资源推荐
- 混部部署、弹性优化 ...

3. 智能重调度

- 碎片最小化
- 节点利用率均衡性、消除热点

4. 南北向生态:

- Spark、Ray、Flink、Serverless
- xGPU、NPU、FPGA

5. 调度持续演进、优化

- 工作流编排: JobFlow
- 资源管理: 精细化资源分配、抢占、算力归一
- 作业性能提升: “应用+资源” 融合调度
-



02

社区角色、治理



社区角色、治理

Role	Requirements	Responsibilities	Privileges
Member	Sponsor from 2 approvers, active in community, contributed to Volcano	Welcome and guide new contributors	Volcano GitHub organization Member
Reivewer	Sponsor from 2 maintainers, has basic experience and knowledge of domain, actively contributed to code and review	Review contributions from community members	Write access to specific packages in relevant repository
Approver	Sponsor from 2 maintainers, has good experience and knowledge of domain, actively contributed to code and review	Review and approve contributions from community members	Write access to specific packages in relevant repository
Maintainer	Sponsor from 2 owners, shown good technical judgement in feature design/development and PR review	Participate in release planning and feature development/maintenance	Top level write access to relevant repository. Name entry in Maintainers file of the repository
Owner	Sponsor from 3 owners, helps drive the overall Volcano project	Drive the overall technical roadmap of the project and set priorities of activities in release planning	Volcano GitHub organization Admin access

Note : It is mandatory for all Volcano community members to follow Volcano [Code of Conduct](#).

社区会议、联络

Regular Community Meeting:

- Asia Time: Friday at 15:00-16:00 (UTC+8) (weekly)

Resources:

- [Meeting notes and agenda](#)
- [Meeting link](#)
- [Meeting Calendar](#) | [Subscribe](#)

Contact:

- [Volcano Slack Channel](#)
- [Mailing List](#)
- [Twitter](#)



03

社区贡献者流程介绍



如何参与社区贡献

- 事先需要准备什么
- 如何寻找合适的issue, 从哪里开始
- 如何创建一个pull request
- 如何进行code review



如何参与社区贡献

➤ 事先需要准备什么

Code of Conduct

https://github.com/volcano-sh/community/blob/master/code_of_conduct.md

<https://www.contributor-covenant.org/version/1/4/code-of-conduct.html>

➤ 开始你的第一个贡献

✓ Find a good issue:

- 熟悉Volcano代码仓库: Volcano, api, devices, resource-exporter, community, website kubegene, kube-state-metrics等
- 点击“issues”，查看所有的issue列表，寻找感兴趣的
- 依据标签搜索新手优化的issue, “help wanted”, “good first issue”等

✓ Work on an issue:

- 通过 “/assign” 将issue分配给自己



如何参与社区贡献

✓ File an issue

- 搜索相关的issue是否已经被登记
- 按照issue要求, 选择类型, 按照模板填写信息, 如问题现象、期待行为、重现步骤、环境信息

Bug Report Report a bug encountered while operating Volcano	Get started
Enhancement Request Suggest an enhancement to the Volcano project	Get started
Report a security vulnerability Please review our security policy for more details.	View policy

✓ Create a PR

- Fork个人仓库, 基于base分支, 创建一个issue分支, base分支通常为master
- 完成自己的代码修改
- 提交commit, 确保commit messages符合格式要求
- Push commit到私有仓库的issue分支.
- 从私有分支, 提交一个pull request 到 [Volcano](#).
- Pull request 必须需要至少两名approver/reviewer的评审

如何参与社区贡献

✓ Code Review

- 遵从代码规范 [good coding guidelines](#)
- 编写规范的commit message [good commit messages](#)
- 拆分大的代码修改为小的patch，方便代码评审和快速迭代
- 给PRs 标记合适的reviewers

✓ Test

- 单元测试：确定函数行为符合预期，保证UT覆盖率
- 集成测试：覆盖组件间的交互符合预期
- 端到端测试：覆盖系统行为符合预期



Join Volcano



Website: <https://volcano.sh/en/>



Github: <https://github.com/volcano-sh/volcano>



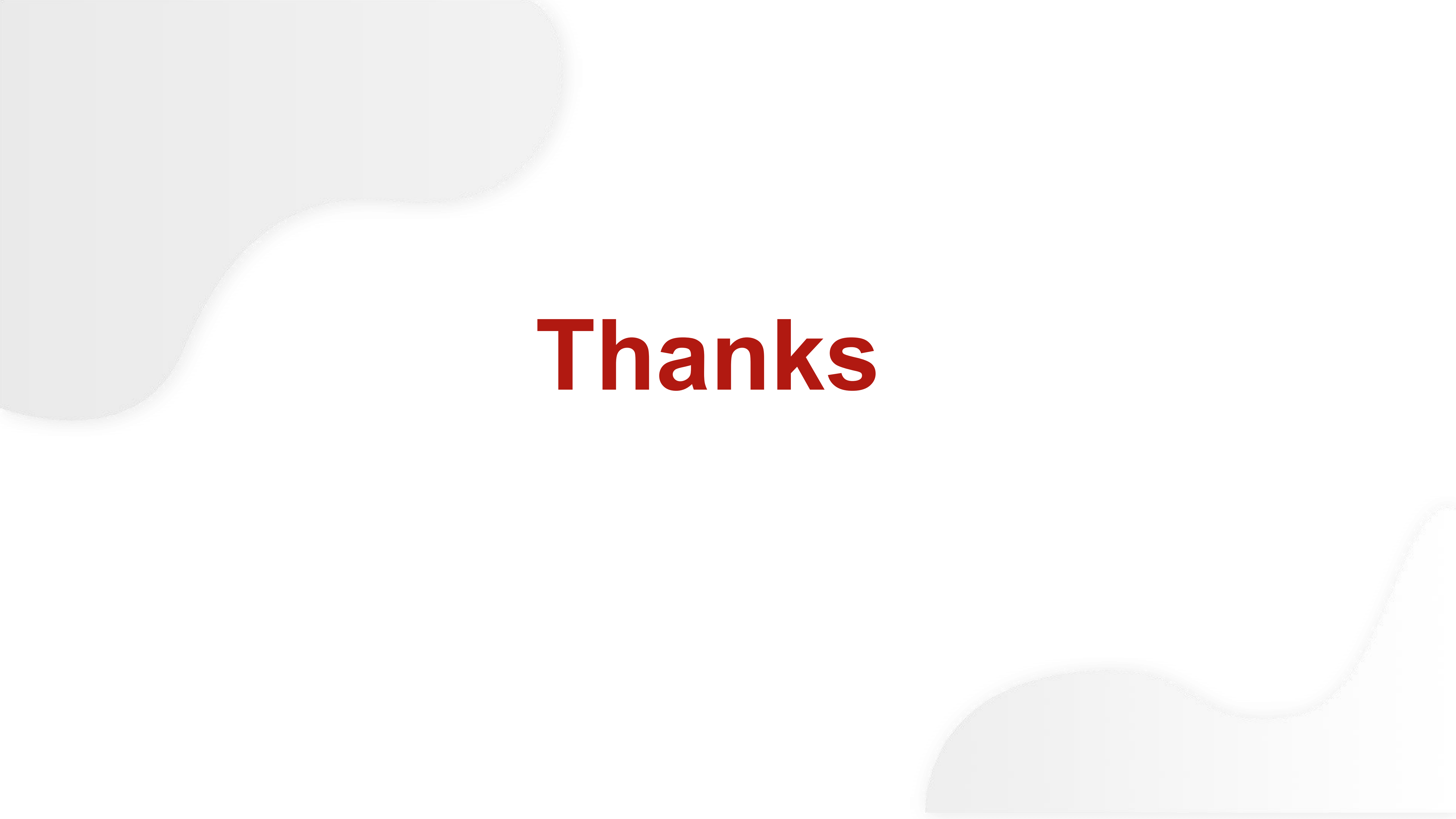
Slack Channel: <https://volcano-sh.slack.com/>



Follow Us



Join Group



Thanks